

Hasnain Asif

AI Engineer · Agentic AI & LLM Systems · Full-Stack Engineer

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📍 Bahawalpur, Pakistan

PROFESSIONAL SUMMARY

AI Engineer with 5+ years of full-stack development experience, specialised in agentic AI systems, RAG pipelines, and LLM integrations deployed on AWS. Built multi-agent LangGraph pipelines, autonomous task execution systems, and enterprise-scale web platforms serving 11,000+ users across 4 countries.

TECHNICAL SKILLS

AI & LLM

Python

LangChain

LangGraph

LlamaIndex

CrewAI

OpenAI Agents SDK

Vector Databases

Large Language Models (LLMs)

Retrieval-Augmented Generation (RAG)

Transformer

Machine Learning

NLP

Prompt Engineering

Embeddings

Multi-Agent Systems

FastAPI

Cloud & DevOps

AWS Lambda

SQS

EventBridge

API Gateway

Bedrock

SageMaker

S3 / ECR

Terraform

GitHub Actions

Docker

CI/CD

Full-Stack

React

Next.js

Node.js

TypeScript

Express.js

Redux

GraphQL

WebSockets

Microservices

REST API

Databases

PostgreSQL

MongoDB

ChromaDB

FAISS

Pinecone

Redis

Semantic Search

BM25

EXPERIENCE

- 🕒 AI Engineer — Agentic AI & Production Systems

Oct 2025 - Present

Self-Employed · Independent AI Engineer

— Built an agentic RAG pipeline with a retrieval evaluator agent that filters documents by relevance before generation — improving answer quality in document-heavy use cases.

— Deployed a distributed multi-agent system on AWS: five Lambda functions (one per agent), S3-backed vector storage, SageMaker embeddings, API Gateway orchestration, and CloudFront + S3 for a Next.js frontend.

— Implemented IaC pipelines using Terraform and GitHub Actions for automated deployment across AWS Bedrock, Lambda, ECR, S3, and CloudFront.
- 🕒 Senior Software Engineer

Oct 2023 - Present

tkxel · Lahore, Pakistan

— Built core scheduling, shift assignment, and timezone systems for an enterprise workforce platform serving 11,000+ users across 170+ franchises in 4 countries — invalid assignments blocked at UI level to prevent payroll errors.

— Engineered a location-anchored DST timezone engine adopted across 50+ components — eliminated production-level shift scheduling errors across the US, Canada, UK, and Australia.

- Led a 5-engineer team to revive a 2-year abandoned multi-tenant social platform with no documentation — owned task breakdown, PR reviews, and all technical decisions.
- Reverse-engineered OpenFGA authorization schema from legacy deployments — restored full role-based permission hierarchy across all admin levels.
- Built and shipped multi-currency Stripe subscriptions (PKR, USD, GBP) on a platform designed for 1M+ audience members across 1,000–5,000 brand Hubs.

🕒 Software Engineer

May 2022 – Oct 2023

Tkxel · Lahore, Pakistan

- Built and shipped enterprise helpdesk features for Tikit, an MS Teams-integrated ticketing platform for high-volume IT, HR, and Finance workflows.
- Built a Custom Views system for an MS Teams-integrated enterprise helpdesk — enabled persistent multi-filter configurations across thousands of concurrent tickets.

🕒 Software Engineer

Oct 2021 – Jan 2022

Invozone · Lahore, Pakistan

- Led frontend delivery on a production multi-chain DEX enabling token swaps across Ethereum, BSC, Polygon, and Avalanche — owned architecture decisions and directed the team end-to-end.
- Integrated MetaMask and Coinbase wallet connections with live on-chain function calls via Web3.js.

🕒 Software Engineer

Aug 2020 – Oct 2021

OptimusFox · Lahore, Pakistan

- Served as tech lead on a token offering (ITO) platform — designed ERD, owned full-stack architecture, and routed tasks across a 4-engineer team.
- Built a blockchain-based tamper-resistant e-voting system with three role-based panels — admin, candidate, and voter — with on-chain ballot execution via Web3.js.
- Owned full-stack delivery across all projects — React + Redux frontends, Node.js/Express backends, PostgreSQL and MongoDB databases.

AI PROJECTS

DocInsight — Verified Document Q&A

LangGraph · LangChain · ChromaDB · OpenAI · FastAPI

A multi-agent RAG system that eliminates manual document review — three specialised agents gate relevance, generate grounded answers, and verify factual support before any response reaches the user.

- Built a 3-agent LangGraph pipeline — Relevance Checker, Research Agent, and Verification Agent — answers verified against source passages before being returned to the user.
- Implemented hybrid retrieval combining BM25 (40%) and ChromaDB vector search (60%) — improves recall on exact-match queries that pure semantic search consistently misses.
- Designed conditional routing so failed verification triggers automatic re-research — unverified answers never surface to the user.

Sidekick — Autonomous AI Work Agent

LangGraph · OpenAI · Playwright · Serper API · Python REPL

A two-model agentic system that executes multi-step tasks across browser, code, files, and search without human supervision — a structurally independent evaluator verifies success criteria are met before any result is returned.

- Built a two-model LangGraph pipeline — worker executes across browser automation, Python REPL, file I/O, and web search, while a separate evaluator independently verifies task completion, eliminating self-evaluation bias.
- Implemented typed feedback loop — structured feedback is injected into the worker system prompt on retry, so the agent knows exactly why it failed before reattempting.

- Built a human-in-the-loop escalation mechanism — when the agent repeats the same mistake after feedback, it surfaces the blocker to the user instead of looping silently.

EDUCATION

BS Software Engineering

University of Engineering & Technology, Taxila

Rawalpindi, Pakistan

CERTIFICATIONS

- **IBM RAG and Agentic AI Specialization**
Coursera · IBM
- **AI Engineer Production Track: Deploy LLMs & Agents at Scale**
Udemy · Ed Donner
- **Agentic Track: The Complete Agent & MCP Course**
Udemy · Ed Donner
- **Serverless Framework Bootcamp: Node.js, AWS & Microservices**
Udemy · Ariel Weinberger

KEY ACHIEVEMENTS

 **11,000+ Users, 4 Countries:** Delivered and maintained an enterprise workforce platform across 170+ franchises in the US, Canada, UK, and Australia without production downtime.

 **Zero Scheduling Errors:** Designed a location-anchored DST timezone engine adopted across 50+ components — eliminated all production-level shift scheduling errors for a globally distributed user base.

 **Revived a Dead Platform:** Led a 5-engineer team to fully restore a 2-year abandoned multi-tenant social platform with no documentation, no infrastructure visibility, and no handoff — shipping live in 4 months.

 **Production Agentic AI:** Built and deployed a distributed multi-agent system on AWS — five independent Lambda agents with S3-backed vector storage, SageMaker embeddings, and full IaC via Terraform.

 **Multi-Currency Payments at Scale:** Shipped Stripe subscription billing in PKR, USD, and GBP on a platform designed for 1M+ audience members across 1,000–5,000 brand Hubs.